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Bearing assembly with load isolation for gas turbine engine

Abstract

A bearing assembly includes an inner race to receive a rotating component having an axis of rotation, and at least one bearing element coupled to the inner race. The at least one bearing element is to rotate with the inner race along the axis of rotation. The bearing assembly includes an outer race coupled to the at least one bearing element, and a bearing housing coupled to the outer race and to be coupled to a static structure. The bearing housing includes a first frangible portion to release upon receipt of a moment load greater than a moment threshold and a second frangible portion to release upon receipt of a radial load greater than a radial threshold.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Indian Provisional Patent Application No. 202311016104, filed Mar. 10, 2023, the entire content of which is incorporated by reference herein.

TECHNICAL FIELD

(2) The present disclosure generally relates to gas turbine engines, and more particularly relates to a bearing assembly with load isolation for a gas turbine engine.

BACKGROUND

(3) Gas turbine engines may be employed to power various devices. For example, a gas turbine engine may be employed to power a mobile platform, such as an aircraft or rotorcraft. Gas turbine engines may include one or more rotating components, which are supported by one or more bearings. Generally, a portion of the bearing is coupled to a static structure of the gas turbine engine. In certain instances, the rotating component may introduce loads into the bearing, which may be transferred to the static structure through the bearing. The transfer of the loads from the bearing to the static structure is undesirable as it may impact the performance of the static structure.

(4) Accordingly, it is desirable to provide a bearing assembly with load isolation for a gas turbine engine, which inhibits the transfer of loads to a static structure associated with the gas turbine engine through the bearing assembly. Furthermore, other desirable features and characteristics of the present disclosure will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and the foregoing technical field and background.

SUMMARY

(5) According to various embodiments, provided is a bearing assembly. The bearing assembly includes an inner race configured to receive a rotating component having an axis of rotation, and at least one bearing element coupled to the inner race. The at least one bearing element is configured to rotate with the inner race along the axis of rotation. The bearing assembly includes an outer race coupled to the at least one bearing element, and a bearing housing coupled to the outer race and configured to be coupled to a static structure. The bearing housing includes a first frangible portion configured to release upon receipt of a moment load greater than a moment threshold and a second frangible portion configured to release upon receipt of a radial load greater than a radial threshold.

(6) The first frangible portion is defined on the bearing housing along an axis substantially perpendicular to the axis of rotation. The second frangible portion is defined on the bearing housing along an axis substantially parallel to the axis of rotation. The first frangible portion includes at least one first groove that defines a reduced thickness. The second frangible portion includes at least one second groove that defines a second reduced thickness. The at least one first groove is continuous about a perimeter of the first frangible portion. The at least one first groove comprises a pair of opposed grooves that cooperate to define the reduced thickness. The first frangible portion includes a plurality of first slots spaced apart about a perimeter of the first frangible portion. The second frangible portion includes a plurality of second slots spaced apart about a second perimeter of the second frangible portion. Each second slot extends along a major axis, and the major axis is substantially parallel to the axis of rotation. Each first slot extends along a major axis, and the major axis is substantially perpendicular to the axis of rotation. The bearing housing includes a first coupling section to be coupled to the outer race, a second coupling section configured to be coupled to the static structure, and the first frangible portion and the second frangible portion are defined between the first coupling section and the second coupling section. The first frangible portion extends radially outward from the first coupling section, and the second frangible portion extends axially from the first frangible portion relative to the axis of rotation. The second frangible portion is coupled to a first end of the second coupling section, and the second coupling section defines a flange at a second end, with the second end opposite the first end. The rotating component is a shaft associated with a gas turbine engine.

(7) Further provided is a bearing assembly for a gas turbine engine. The bearing assembly includes an inner race configured to receive a rotating component having an axis of rotation. The bearing assembly includes at least one bearing element coupled to the inner race, and the at least one bearing element is configured to rotate with the inner race along the axis of rotation. The bearing assembly includes an outer race coupled to the at least one bearing element, and a bearing housing coupled to the outer race and configured to be coupled to a static structure. The bearing housing includes a first frangible portion and a second frangible portion. The first frangible portion extends along a first axis substantially perpendicular to the axis of rotation, and the first frangible portion is configured to release upon receipt of a moment load greater than a moment threshold. The second frangible portion extends along a second axis substantially parallel to the axis of rotation and the second frangible portion is configured to release upon receipt of a radial load greater than a radial threshold.
(8) The first frangible portion includes a pair of opposed grooves that cooperate to define a reduced thickness and each groove of the pair of opposed grooves is continuous about a perimeter of the first frangible portion. The first frangible portion includes a plurality of first slots spaced apart about a perimeter of the first frangible portion and the second frangible portion includes a plurality of second slots spaced apart about a second perimeter of the second frangible portion. Each second slot extends along a major axis, the major axis is substantially parallel to the axis of rotation. Each first slot extends along a major axis, and the major axis is substantially perpendicular to the axis of rotation.

Description

DESCRIPTION OF THE DRAWINGS
(1) The exemplary embodiments will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:
(2) FIG. **1** is a schematic cross-sectional illustration of a gas turbine engine, which includes an exemplary bearing assembly with load isolation in accordance with the various teachings of the present disclosure;
(3) FIG. **2** is a detail cross-sectional view, taken at **2** on FIG. **1**, of the bearing assembly with load isolation coupled about a shaft associated with the gas turbine engine in accordance with the various teachings of the present disclosure;
(4) FIG. **3** is a perspective view of an exemplary bearing housing associated with the bearing assembly;
(5) FIG. **4** is a cross-sectional view of the bearing housing of FIG. **3** taken along line **4-4** of FIG. **3**;
(6) FIG. **5** is a cross-sectional view of the bearing housing of FIG. **3** taken along line **5-5** of FIG. **3**;
(7) FIG. **6** is another exemplary bearing housing for use with the bearing assembly of FIG. **1**;
(8) FIG. **7** is a cross-sectional view of the bearing housing of FIG. **6** taken along line **7-7** of FIG. **6**;
(9) FIG. **8** is yet another exemplary bearing housing for use with the bearing assembly of FIG. **1**; and
(10) FIG. **9** is a cross-sectional view of the bearing housing of FIG. **8** taken along line **9-9** of FIG. **8**.
DETAILED DESCRIPTION
(11) The following detailed description is merely exemplary in nature and is not intended to limit the application and uses. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with any type of engine that would benefit from a bearing assembly with load isolation and the gas turbine engine described herein is

merely one exemplary embodiment according to the present disclosure. In addition, while the bearing assembly is described herein as being used with a gas turbine engine onboard a mobile platform, such as a bus, motorcycle, train, motor vehicle, marine vessel, aircraft, rotorcraft and the like, the various teachings of the present disclosure can be used with a gas turbine engine on a stationary platform. Further, it should be noted that many alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure. In addition, while the figures shown herein depict an example with certain arrangements of elements, additional intervening elements, devices, features, or components may be present in an actual embodiment. It should also be understood that the drawings are merely illustrative and may not be drawn to scale.

(12) As used herein, the term “axial” refers to a direction that is generally parallel to or coincident with an axis of rotation, axis of symmetry, or centerline of a component or components. For example, in a cylinder or disc with a centerline and generally circular ends or opposing faces, the “axial” direction may refer to the direction that generally extends in parallel to the centerline between the opposite ends or faces. In certain instances, the term “axial” may be utilized with respect to components that are not cylindrical (or otherwise radially symmetric). For example, the “axial” direction for a rectangular housing containing a rotating shaft may be viewed as a direction that is generally parallel to or coincident with the rotational axis of the shaft. Furthermore, the term “radially” as used herein may refer to a direction or a relationship of components with respect to a line extending outward from a shared centerline, axis, or similar reference, for example in a plane of a cylinder or disc that is perpendicular to the centerline or axis. In certain instances, components may be viewed as “radially” aligned even though one or both of the components may not be cylindrical (or otherwise radially symmetric). Furthermore, the terms “axial” and “radial” (and any derivatives) may encompass directional relationships that are other than precisely aligned with (e.g., oblique to) the true axial and radial dimensions, provided the relationship is predominantly in the respective nominal axial or radial direction. As used herein, the term “about” denotes within 10% to account for manufacturing tolerances. In addition, the term “substantially” denotes within 10% to account for manufacturing tolerances.

(13) With reference to FIG. **1**, a simplified cross-sectional view of an exemplary gas turbine engine **100** is shown with the remaining portion of the gas turbine engine **100** being substantially axisymmetric about an axis of rotation **140**, which also comprises a longitudinal axis for the gas turbine engine **100**. As will be discussed herein, the gas turbine engine **100** includes a bearing assembly **200** with load isolation, which inhibits loads, such as shear loads and moment loads, from passing through the bearing assembly **200** into a surrounding fixed or static structure. In the depicted embodiment, the gas turbine engine **100** is an annular multi-spool turbofan gas turbine jet engine for use with a vehicle, such as an aircraft **99**, although other arrangements and uses may be provided. For example, in other embodiments, the gas turbine engine **100** may assume the form of a non-propulsive engine, such as an Auxiliary Power Unit (APU) deployed onboard the aircraft **99**, or an industrial power generator.

(14) In this example, the gas turbine engine **100** includes a fan section **102**, a compressor section **104**, a combustor section **106**, a turbine section **108**, and an exhaust section **110**. The fan section **102** includes a fan **112** that draws air into the gas turbine engine **100** and accelerates it. A fraction of the accelerated air exhausted from the fan **112** is directed through an outer (or first) bypass duct **116** and the remaining fraction of air exhausted from the fan **112** is directed into the compressor section **104**. In the embodiment of FIG. **1**, the compressor section **104** includes one or more compressors **120**, and in one example, includes four compressors **120**. However, in other embodiments, the number of compressors **120** in the compressor section **104** may vary. In the depicted embodiment, the compressors **120** sequentially raise the pressure of the air and direct a majority of the high pressure air into the combustor section **106**. A fraction of the compressed air bypasses the combustor section **106** and is used to cool, among other components, turbine blades in

the turbine section **108**.

(15) In the embodiment of FIG. **1**, in the combustor section **106**, which includes a combustion chamber **124**, the high pressure air is mixed with fuel, which is combusted. The high-temperature combustion air is directed into the turbine section **108**. In this example, the turbine section **108** includes one or more turbines **126**, and in one example, includes four turbines **126** disposed in axial flow series. However, it will be appreciated that the number of turbines, and/or the configurations thereof, may vary. In this embodiment, the high-temperature air from the combustor section **106** expands through and rotates each turbine **126**. As the turbines **126** rotate, each drives equipment in the gas turbine engine **100** via at least one concentrically disposed spool or shaft **130**. The rotation of the spool or shaft **130** provides power output, which may be utilized in a variety of different manners, depending upon whether the gas turbine engine **100** assumes the form of a turbofan, turboprop, turboshaft, turbojet engine, or an auxiliary power unit, to list but a few examples. The spool or shaft **130** rotates about the axis of rotation **140**.

(16) In this example, the bearing assembly **200** is coupled to a rotating component having an axis of rotation, such as the shaft **130** having the axis of rotation **140**. The bearing assembly **200** is coupled to the shaft **130** proximate, downstream of or behind the fan **112** in the fan section **102**. The bearing assembly **200** supports the shaft **130** for rotation. It should be noted that in other embodiments, the bearing assembly **200** may be used to support rotation of other rotating components associated with the gas turbine engine **100** and the location of the bearing assembly **200** is merely an example. With reference to FIG. **2**, a detail cross-sectional view of a portion of the fan section **102** is shown. In one example, the bearing assembly **200** includes an inner race **202**, one or more bearing elements **204**, an outer race **206** and a bearing housing **208**. In this example, the inner race **202** is coupled to the shaft **130**, and the bearing housing **208** is coupled to a static structure **210**, which in one example, is a flange **212** that is coupled to a strut **214** associated with the compressor section **104** (FIG. **1**). The bearing housing **208** is coupled to the flange **212** via one or more mechanical fasteners, such as bolts **216**. It should be noted that while not illustrated herein, the bearing assembly **200** may also include one or more seals.

(17) As discussed, the inner race **202** is coupled to the shaft **130** via a press-fit, for example, but swaging and other techniques may be used to couple the inner race **202** to the shaft **130**. The inner race **202** is coupled of a metal or metal alloy, and is cast, forged, stamped, additively manufactured, etc. The inner race **202** is annular, and is sized to be coupled about the diameter of the shaft **130** to support the shaft **130** for rotation relative to the outer race **206** and the static structure **210**. The inner race **202** has a first inner surface **220** opposite a second inner surface **222**. The first inner surface **220** is coupled to the shaft **130**, and the second inner surface **222** is coupled to the bearing elements **204**. In one example, the second inner surface **222** may include a concave inner recess **224**, which is shaped to receive and partially retain the bearing elements **204**. The second inner surface **222** defines an inner perimeter or diameter of the inner race **202**, and surrounds a bore defined through the inner race **202**.

(18) The bearing elements **204** enable the rotation of the inner race **202** relative to the outer race **206**. In one example, the bearing elements **204** comprise a plurality of ball bearings. In other examples, the bearing elements **204** may comprise a plurality of roller bearings, cylindrical roller bearings, needle bearings, etc. The bearing elements **204** are composed of a metal or metal alloy, and may be machined, additively manufactured, etc. The bearing elements **204** may be arranged between the inner race **202** and the outer race **206**, and retained by the inner recess **224** and a concave outer recess **226** of the outer race **206**. In other embodiments, the bearing elements **204** may be supported in a cage.

(19) The outer race **206** is coupled to the bearing housing **208** via a press-fit, for example, but other techniques may be used, such as welding, mechanical fasteners, etc. The outer race **206** is coupled of a metal or metal alloy, and is cast, forged, stamped, additively manufactured, etc. The outer race **206** is annular, and is sized to be coupled about the bearing elements **204** and the inner race **202**.

The outer race **206** has a first outer surface **230** opposite a second outer surface **232**. The first outer surface **230** is coupled to the bearing housing **208**, and the second outer surface **232** is coupled to the bearing elements **204**. In one example, the second outer surface **232** includes the outer recess **226**, which is shaped to receive and partially retain the bearing elements **204**. The bearing elements **204** rotate relative to the outer race **206**. The second outer surface **232** defines an inner perimeter or diameter of the outer race **206**, and surrounds a bore defined through the outer race **206**.

(20) The bearing housing **208** is coupled to the outer race **206**, and to the flange **212** associated with the static structure **210**. The bearing housing **208** is composed of a metal or metal alloy, and is cast, forged, stamped, machined, additively manufactured, etc. The bearing housing **208** includes a first coupling section or race coupling section **240**, a second coupling section or static coupling section **242** and a frangible section **244** between the race coupling section **240** and the static coupling section **242**. In this example, the bearing housing **208** is substantially symmetric about the axis of rotation **140**. The race coupling section **240** is at a first housing end **246**, and the static coupling section **242** extends from the frangible section **244** to a second housing end **248**. The second housing end **248** is opposite the first housing end **246**.

(21) With reference to FIG. **3**, the bearing housing **208** is shown in greater detail. The race coupling section **240** is cylindrical, and is sized to be positioned about the outer race **206**, the bearing elements **204** and the inner race **202**. With reference to FIG. **4**, the race coupling section **240** includes an outer race surface **250** opposite an inner race surface **252**. The outer race surface **250** forms an outer surface or outer perimeter of the race coupling section **240**, and the inner race surface **252** is coupled to the outer race **206** via press-fit, welding, mechanical fasteners. etc. The inner race surface **252** defines an inner perimeter or diameter of the race coupling section **240**, and surrounds a bore defined through the race coupling section **240**. The race coupling section **240** also has a first race end **254** opposite a second race end **256**. The first race end **254** is defined at the first housing end **246** and the second race end **256** is coupled to the frangible section **244**. With additional reference to FIG. **5**, the race coupling section **240** generally extends for a first distance D**1** from the first housing end **246** to the frangible section **244** along the axis of rotation **140**. In one example, the first distance D**1** is about 0.8 inches (in.) to about 1.2 inches (in.). The race coupling section **240** has a first thickness T**1** defined between the outer race surface **250** and the inner race surface **252**. The first thickness T**1** is uniform over the race coupling section **240**. In one example, the first thickness T**1** is about 0.1 inches (in.) to about 0.5 inches (in.).

(22) With reference back to FIG. **3**, the static coupling section **242** is frustoconical, and is sized to extend from the frangible section **244** to couple to the flange **212** associated with the static structure **210** (FIG. **2**). In this example, the static coupling section **242** tapers from the second housing end **248** to the frangible section **244**. With reference to FIG. **4**, the static coupling section **242** includes an outer static surface **260** opposite an inner static surface **262**. The outer static surface **260** forms an outer surface or outer perimeter of the static coupling section **242**, and the inner static surface **262** faces the shaft **130**. The inner static surface **262** defines an inner perimeter or diameter of the static coupling section **242**, and surrounds a bore defined through the static coupling section **242**. The static coupling section **242** also has a first static end **264** opposite a second static end **266**. The first static end **264** is coupled to the frangible section **244**, and the second static end **266** is defined at the second housing end **248**. The second static end **266** includes a flange **268**, which extends radially about the perimeter or circumference of the static coupling section **242** at the second static end **266**. The flange **268** may include one or more bores **269**, which receive the one or more bolts **216** (FIG. **1**) to couple the bearing housing **208** to the static structure **210**. With additional reference to FIG. **5**, the static coupling section **242** generally extends for a second distance D**2** from the frangible section **244** to the second housing end **248** along the axis of rotation **140**. In one example, the second distance D**2** is about 3.0 inches (in.) to about 3.5 inches (in.). The second distance D**2** is different and greater than the distance D**1**. The static coupling section **242** has a second thickness T**2** defined between the outer static surface **260** and the inner static surface **262**. The second

thickness T**2** is uniform over the static coupling section **242**. The second thickness T**2** is about the same as the thickness T**1**.

(23) With reference back to FIGS. **3** and **4**, the frangible section **244** is defined between the race coupling section **240** and the static coupling section **242**. The frangible section **244** is substantially L-shaped, and includes a first frangible portion **270** and a second frangible portion **272**. With reference to FIG. **5**, the first frangible portion **270** extends along an axis A**1**, which is substantially perpendicular to the axis of rotation **140**. Thus, generally, the first frangible portion **270** extends radially from and relative to the race coupling section **240**. The first frangible portion **270** includes a first outer surface **274** opposite a first inner surface **276**. The first outer surface **274** defines an outer surface or outer perimeter of the first frangible portion **270**, and the first inner surface **276** defines an inner surface or perimeter of the first frangible portion **270**. The first frangible portion **270** also has a first portion end **278** opposite a second portion end **280**. The first portion end **278** is coupled to the second race end **256** and the second portion end **280** is coupled to the second frangible portion **272**. The first frangible portion **270** generally extends for a third distance D**3** from the second race end **256** to the second frangible portion **272** in a radial direction relative to the axis of rotation **140**. In one example, the third distance D**3** is about 0.8 inches (in.) to about 1.2 inches (in.). The first frangible portion **270** has a third thickness T**3** at the first portion end **278** and the second portion end **280**, and has a fourth thickness T**4** defined between the first portion end **278** and the second portion end **280**. The third thickness T**3** is about the same as the first thickness T**1** and the second thickness T**2**. The fourth thickness T**4** is different and less than the third thickness T**3**, and thus, the first thickness T**1** and the second thickness T**2**. In one example, the fourth thickness T**4** is about 0.03 inches (in.) to about 0.08 inches (in.). The fourth thickness T**4** defines a first area of reduced thickness **282**. In this example, the first area of reduced thickness **282** extends for a first length L**1** about the perimeter of the bearing housing **208**. In one example, the first length L**1** is about 0.2 inches (in.) to about 0.5 inches (in.).

(24) In this example, the first area of reduced thickness **282** is defined by a first concave groove **284** recessed into the first outer surface **274** and an opposite second concave groove **286** recessed into the first inner surface **276**. The opposed grooves **284**, **286** reduce the thickness of the first frangible portion **270** to the fourth thickness T**4** between the first portion end **278** and the second portion end **280**. By reducing the thickness of the first frangible portion **270** to the fourth thickness T**4**, the first frangible portion **270** is configured to release, break, or fracture along the first area of reduced thickness **282** to inhibit the transfer of a moment load that is greater than a predetermined moment threshold. In one example, the moment threshold is about 70,000 pound-force inch (lbf-in.) to about 90,000 pound-force inch (lbf-in.). By releasing, breaking or fracturing along the first area of reduced thickness **282** upon receipt of a moment load greater than the moment threshold, the first frangible portion **270** isolates the moment load from the static structure **210** (FIG. **2**). This inhibits the static structure **210** from absorbing the moment load.

(25) The second frangible portion **272** extends along an axis A**2**, which is substantially parallel to the axis of rotation **140**. Thus, generally, the second frangible portion **272** extends axially relative to the race coupling section **240**. The second frangible portion **272** includes a second outer surface **290** opposite a second inner surface **292**. The second outer surface **290** defines an outer surface or outer perimeter of the second frangible portion **272**, and the second inner surface **292** defines an inner surface or perimeter of the second frangible portion **272**. The second frangible portion **272** also has a third portion end **294** opposite a fourth portion end **296**. The third portion end **294** is coupled to the first frangible portion **270** and the fourth portion end **296** is coupled to the static coupling section **242**. The second frangible portion **272** generally extends for a fourth distance D**4** from the first frangible portion **270** to the static coupling section **242** along the axis of rotation **140**. In one example, the fourth distance D**4** is about 0.8 inches (in.) to about 1.2 inches (in.). The second frangible portion **272** has a fifth thickness T**5** at the third portion end **294** and the fourth portion end **296**, and has a sixth thickness T**6** defined between the third portion end **294** and the

fourth portion end **296**. The fifth thickness T**5** is about the same as the first thickness T**1**, the second thickness T**2** and the third thickness T**3**. The sixth thickness T**6** is different and less than the first thickness T**1**, the second thickness T**2** and the third thickness T**3**. In one example, the sixth thickness T**6** is the about the same as the fourth thickness T**4**. The sixth thickness T**6** defines a second area of reduced thickness **298**. In this example, the second area of reduced thickness **298** extends for a second length L**2** about the perimeter of the bearing housing **208**. In one example, the second length L**2** is about the same as the first length L**1**. The first area of reduced thickness **282** is spaced a distance apart from the second area of reduced thickness **298**.

(26) In this example, the second area of reduced thickness **298** is defined by a third concave groove **300** recessed into the second outer surface **290** and an opposite fourth concave groove **302** recessed into the second inner surface **292**. The opposed grooves **300**, **302** reduce the thickness of the second frangible portion **272** to the sixth thickness T**6** between the third portion end **294** and the fourth portion end **296**. By reducing the thickness of the second frangible portion **272** to the sixth thickness T**6**, the second frangible portion **272** is configured to release, break or fracture along the second area of reduced thickness **298** to inhibit the transfer of a shear or radial load that is greater than a predetermined shear or radial threshold. In one example, the radial threshold is about 45,000 pound-force (lbf) to about 60,000 pound-force (lbf). By releasing, breaking or fracturing along the second area of reduced thickness **298** upon receipt of a radial load greater than the radial threshold, the second frangible portion **272** isolates the radial load from the static structure **210** (FIG. **2**). This inhibits the static structure **210** from absorbing the radial load.

(27) In this example, the grooves **284**, **286**, **300**, **302** are defined to extend about the circumference of the bearing housing **208** on the respective one of the surfaces **274**, **276**, **290**, **292**, however, in other examples, one of the grooves **284**, **286**, **300**, **302** may be defined to a greater depth on one of the surfaces **274**, **276**, **290**, **292** such that the first frangible portion **270** and the second frangible portion **272** have the respective one of the fourth thickness T**4** and the sixth thickness T**6** with a single groove. In addition, while the areas of reduced thickness **282**, **298** are described as areas, the grooves **284**, **286**, **300**, **302** may be defined to result in a line of reduced thickness, if desired.

(28) In addition, it should be noted that while the first frangible portion **270** and the second frangible portion **272** of the bearing housing **208** are shown in FIGS. **1-5** as comprising continuous grooves **284**, **286**, **300**, **302** defined about the perimeter or circumference, in other embodiments, the first frangible portion **270** and the second frangible portion **272** may be configured differently to release, fracture or break upon receipt of a respective one of a predetermined moment load and radial load. For example, with reference to FIG. **6**, a bearing housing **408** is shown for use with the gas turbine engine **100** (FIG. **1**). As the bearing housing **408** includes components that are the same or similar to components of the bearing housing **208** discussed with regard to FIGS. **1-5**, the same reference numerals will be used to denote the same or similar components. The bearing housing **408** is coupled to the outer race **206**, and to the flange **212** associated with the static structure **210** (FIG. **2**). The bearing housing **408** is composed of a metal or metal alloy, and is cast, forged, stamped, machined, additively manufactured, etc. In this example, the bearing housing **408** includes the first coupling section or race coupling section **240**, the second coupling section or static coupling section **242** and a frangible section **444** between the race coupling section **240** and the static coupling section **242**. In this example, the bearing housing **408** is substantially symmetric about the axis of rotation **140**. The race coupling section **240** is at the first housing end **246**, and the static coupling section **242** extends from the frangible section **444** to the second housing end **248**. The second housing end **248** is opposite the first housing end **246**.

(29) The race coupling section **240** is cylindrical, and is sized to be positioned about the outer race **206**, the bearing elements **204** and the inner race **202** (FIG. **2**). With reference to FIG. **7**, the race coupling section **240** has the first thickness T**1** defined between the outer race surface **250** and the inner race surface **252**. The static coupling section **242** is frustoconical, and is sized to extend from the frangible section **444** to couple to the flange **212** associated with the static structure **210** (FIG.

2). The static coupling section **242** has the second thickness T**2** defined between the outer static surface **260** and the inner static surface **262**.

(30) The frangible section **444** is defined between the race coupling section **240** and the static coupling section **242**. The frangible section **444** is substantially L-shaped, and includes a first frangible portion **470** and a second frangible portion **472**. The first frangible portion **470** extends along the axis A**1**, which is substantially perpendicular to the axis of rotation **140**. Thus, generally, the first frangible portion **470** extends radially from and relative to the race coupling section **240**. The first frangible portion **470** includes a first outer surface **474** opposite a first inner surface **476**. The first outer surface **474** defines an outer surface or outer perimeter of the first frangible portion **470**, and the first inner surface **476** defines an inner surface or perimeter of the first frangible portion **470**. The first frangible portion **470** also has a first portion end **478** opposite a second portion end **480**. The first portion end **478** is coupled to the second race end **256** and the second portion end **480** is coupled to the second frangible portion **472**. The first frangible portion **470** generally extends for the third distance D**3** from the second race end **256** to the second frangible portion **472** in a radial direction relative to the axis of rotation **140**. The first frangible portion **470** has the third thickness T**3** at the first portion end **478** and the second portion end **480**.

(31) In this example, with reference back to FIG. **6**, the first frangible portion **470** includes a plurality of first slots **482** defined between the first portion end **478** and the second portion end **480** about the perimeter of the first frangible portion **470**. Each of first slots **482** are substantially oval in shape, and are spaced apart about the perimeter of the first frangible portion **470**. Each of the first slots **482** have a major axis M**1**, which is substantially perpendicular to the axis of rotation **140**. The first slots **482** are defined through the first outer surface **474** and the first inner surface **476** such that the first slots **482** result in a void or an area of the first frangible portion **470** that is removed. The formation of the first slots **482** about the perimeter of the first frangible portion **470** also results in a plurality of first ligaments **484** being defined about the perimeter of the first frangible portion **470**. A respective first ligament **484** is defined between each of the first slots **482**. Each of the first ligaments **484** is configured to release, break or fracture to inhibit the transfer of the moment load that is greater than the predetermined moment threshold. By releasing, breaking or fracturing along the first ligaments **484** upon receipt of the moment load greater than the moment threshold, the first frangible portion **470** isolates the moment load from the static structure **210** (FIG. **2**). This inhibits the static structure **210** from absorbing the moment load.

(32) With reference to FIG. **7**, the second frangible portion **472** extends along the axis A**2**, which is substantially parallel to the axis of rotation **140**. Thus, generally, the second frangible portion **472** extends axially relative to the race coupling section **240**. The second frangible portion **472** includes a second outer surface **490** opposite a second inner surface **492**. The second outer surface **490** defines an outer surface or outer perimeter of the second frangible portion **472**, and the second inner surface **492** defines an inner surface or perimeter of the second frangible portion **472**. The second frangible portion **472** also has a third portion end **494** opposite a fourth portion end **496**. The third portion end **494** is coupled to the first frangible portion **470** and the fourth portion end **496** is coupled to the static coupling section **242**. The second frangible portion **472** generally extends for the fourth distance D**4** from the first frangible portion **470** to the static coupling section **242** relative to the axis of rotation **140**. The second frangible portion **472** has the fifth thickness T**5** at the third portion end **494** and the fourth portion end **496**.

(33) In this example, the second frangible portion **472** includes a plurality of second slots **498** defined between the third portion end **494** and the fourth portion end **496** about the perimeter of the second frangible portion **472**. Each of second slots **498** are substantially oval in shape, and are spaced apart about the perimeter of the second frangible portion **472**. Each of the second slots **498** have a second major axis M**2**, which is substantially parallel to the axis of rotation **140**. The second slots **498** are spaced a distance apart from the first slots **482**. The second slots **498** are defined through the second outer surface **490** and the second inner surface **492** such that the second slots

498 result in a void or an area of the second frangible portion **472** that is removed. The formation of the second slots **498** about the perimeter of the second frangible portion **472** also results in a plurality of second ligaments **500** being defined about the perimeter of the second frangible portion **472**. A respective second ligament **500** is defined between each of the second slots **498**. Each of the second ligaments **500** is configured to release, break or fracture to inhibit the transfer of the shear or radial load that is greater than the predetermined radial threshold. By releasing, breaking or fracturing along the second ligaments **500** upon receipt of the radial load greater than the radial threshold, the second frangible portion **472** isolates the radial load from the static structure **210** (FIG. **2**). This inhibits the static structure **210** from absorbing the radial load.
(34) In addition, it should be noted that while the first frangible portion **270** and the second frangible portion **272** of the bearing housing **208** are shown in FIGS. **1-5** as comprising continuous grooves **284**, **286**, **300**, **302** defined about the perimeter or circumference, in other embodiments, the first frangible portion **270** and the second frangible portion **272** may be configured differently to release, fracture or break upon receipt of a respective one of a predetermined moment load and radial load. For example, with reference to FIG. **8**, a bearing housing **608** is shown for use with the gas turbine engine **100** (FIG. **1**). As the bearing housing **608** includes components that are the same or similar to components of the bearing housing **208** discussed with regard to FIGS. **1-5**, the same reference numerals will be used to denote the same or similar components. The bearing housing **608** is coupled to the outer race **206**, and to the flange **212** associated with the static structure **210** (FIG. **2**). The bearing housing **608** is composed of a metal or metal alloy, and is cast, forged, stamped, machined, additively manufactured, etc. In this example, the bearing housing **608** includes the first coupling section or race coupling section **240**, the second coupling section or static coupling section **242** and a frangible section **644** between the race coupling section **240** and the static coupling section **242**. In this example, the bearing housing **608** is substantially symmetric about the axis of rotation **140**. The race coupling section **240** is at the first housing end **246**, and the static coupling section **242** extends from the frangible section **644** to the second housing end **248**. The second housing end **248** is opposite the first housing end **246**.
(35) The race coupling section **240** is cylindrical, and is sized to be positioned about the outer race **206**, the bearing elements **204** and the inner race **202** (FIG. **2**). With reference to FIG. **9**, the race coupling section **240** has the first thickness T**1** defined between the outer race surface **250** and the inner race surface **252**. The static coupling section **242** is frustoconical, and is sized to extend from the frangible section **444** to couple to the flange **212** associated with the static structure **210** (FIG. **2**). The static coupling section **242** has the second thickness T**2** defined between the outer static surface **260** and the inner static surface **262**.
(36) The frangible section **644** is defined between the race coupling section **240** and the static coupling section **242**. The frangible section **644** is substantially L-shaped, and includes a first frangible portion **670** and a second frangible portion **672**. The first frangible portion **670** extends along the axis A**1**, which is substantially perpendicular to the axis of rotation **140**. Thus, generally, the first frangible portion **670** extends radially from and relative to the race coupling section **240**. The first frangible portion **670** includes a first outer surface **674** opposite a first inner surface **676**. The first outer surface **674** defines an outer surface or outer perimeter of the first frangible portion **670**, and the first inner surface **676** defines an inner surface or perimeter of the first frangible portion **670**. The first frangible portion **670** also has a first portion end **678** opposite a second portion end **680**. The first portion end **678** is coupled to the second race end **256** and the second portion end **680** is coupled to the second frangible portion **672**. The first frangible portion **670** generally extends for the third distance D**3** from the second race end **256** to the second frangible portion **672** in a radial direction relative to the axis of rotation **140**. The first frangible portion **670** has the third thickness T**3** at the first portion end **678** and the second portion end **680**.
(37) In this example, with reference back to FIG. **8**, the first frangible portion **670** includes a plurality of first slots **682** defined between the first portion end **678** and the second portion end **680**

about the perimeter of the first frangible portion **670**. Each of first slots **682** are substantially oval in shape, and are spaced apart about the perimeter of the first frangible portion **470**. Each of the first slots **682** have a major axis M**1**, which is substantially perpendicular to the axis of rotation **140**. The first slots **682** are defined through the first outer surface **674** and the first inner surface **676** such that the first slots **682** result in a void or an area of the first frangible portion **670** that is removed. The formation of the first slots **682** about the perimeter of the first frangible portion **670** also results in a plurality of first ligaments **684** being defined about the perimeter of the first frangible portion **670**. A respective first ligament **684** is defined between each of the first slots **682**. Each of the first ligaments **684** is configured to release, break or fracture to inhibit the transfer of the moment load that is greater than the predetermined moment threshold. By releasing, breaking, or fracturing along the first ligaments **684** of the first frangible portion **670** upon receipt of the moment load greater than the moment threshold, the first frangible portion **670** isolates the moment load from the static structure **210** (FIG. **2**). This inhibits the static structure **210** from absorbing the moment load.

(38) With reference to FIG. **9**, the second frangible portion **672** extends along the axis A**2**, which is substantially parallel to the axis of rotation **140**. Thus, generally, the second frangible portion **672** extends axially relative to the race coupling section **240**. The second frangible portion **672** includes a second outer surface **690** opposite a second inner surface **692**. The second outer surface **690** defines an outer surface or outer perimeter of the second frangible portion **672**, and the second inner surface **692** defines an inner surface or perimeter of the second frangible portion **672**. The second frangible portion **672** also has a third portion end **694** opposite a fourth portion end **696**. The third portion end **694** is coupled to the first frangible portion **670** and the fourth portion end **696** is coupled to the static coupling section **242**. The second frangible portion **672** generally extends for the fourth distance D**4** from the first frangible portion **670** to the static coupling section **242** relative to the axis of rotation **140**. The second frangible portion **672** has the fifth thickness T**5** at the third portion end **694** and the fourth portion end **696**.

(39) In this example, the second frangible portion **672** includes a plurality of second slots **698** defined between the third portion end **694** and the fourth portion end **696** about the perimeter of the second frangible portion **672**. Each of second slots **698** are substantially oval in shape, and are spaced apart about the perimeter of the second frangible portion **672**. Each of the second slots **698** have a second major axis M**2**, which is substantially perpendicular to the axis of rotation **140**. The second slots **698** are spaced a distance apart from the first slots **682**. The second slots **698** are defined through the second outer surface **690** and the second inner surface **692** such that the second slots **698** result in a void or an area of the second frangible portion **672** that is removed. The formation of the second slots **698** about the perimeter of the second frangible portion **672** also results in a plurality of second ligaments **700** being defined about the perimeter of the second frangible portion **672**. A respective second ligament **700** is defined between each of the second slots **598**. Each of the second ligaments **700** is configured to release, break or fracture to inhibit the transfer of the shear or radial load that is greater than the predetermined radial threshold. By releasing, breaking or fracturing along the second ligaments **700** upon receipt of the radial load greater than the radial threshold, the second frangible portion **672** isolates the radial load from the static structure **210** (FIG. **2**). This inhibits the static structure **210** from absorbing the radial load.

(40) It should be noted that while the bearing housing **208**, **408**, **608** is illustrated herein as including the grooves **284**, **286**, **300**, **302**, the slots **482**, **498** and the slots **682**, **698**, respectively, a bearing housing may include any combination of the grooves **284**, **286**, **300**, **302**, the slots **482**, **498** and the slots **682**, **698** to enable the bearing housing to isolate moment and radial loads from the static structure **210** (FIG. **2**).

(41) In one example, with reference to FIGS. **2-9**, the inner race **202**, the bearing elements **204** and the outer race **206** are coupled to the shaft **130**. With the bearing housing **208**, **408**, **608** formed, the bearing housing **208**, **408**, **608** is coupled to the outer race **206** such that the outer race **206** is

received within and coupled to the race coupling section **240**. With the outer race **206** coupled to the bearing housing **208**, **408**, **608**, the flange **268** is coupled to the static structure **210** via the bolts **216**. With the bearing housing **208**, **408**, **608** installed in the gas turbine engine **100** (FIG. **1**), as the gas turbine engine **100** operates, the shaft **130** rotates.
(42) In the instance of a moment load being imparted to the bearing assembly **200** that is greater than the moment threshold, the first frangible portion **270**, **470**, **670** releases, breaks or fractures to absorb the load and isolate the static structure **210** from the moment load. In the example of the first frangible portion **270**, the first frangible portion **270** releases, breaks or factures along the grooves **284**, **286** upon receipt of the moment load that is greater than the moment threshold. In the example of the first frangible portion **470**, **670**, the first frangible portion **470**, **670** releases, breaks or factures along the first ligaments **484**, **684** upon receipt of the moment load that is greater than the moment threshold.
(43) In the instance of a shear or radial load being imparted to the bearing assembly **200** that is greater than the radial threshold, the second frangible portion **272**, **472**, **672** releases, breaks or fractures to absorb the load and isolate the static structure **210** from the radial load. In the example of the second frangible portion **272**, the second frangible portion **272** releases, breaks or factures along the grooves **300**, **302** upon receipt of the radial load that is greater than the radial threshold. In the example of the second frangible portion **472**, **672**, the second frangible portion **472**, **672** releases, breaks or factures along the second ligaments **500**, **700** upon receipt of the radial load that is greater than the radial threshold.
(44) Thus, the bearing assembly **200** including the bearing housing **208**, **408**, **608** provides load isolation for the gas turbine engine **100** (FIG. **1**), which inhibits the transfer of moment and radial loads to the static structure **210** through the bearing assembly **200**. Each of the first frangible portion **270**, **470**, **670** and the second frangible portion **272**, **472**, **672** may release, break or fracture independently upon receipt of the respective moment load or radial load that exceeds the respective moment threshold or radial threshold. This inhibits the static structure **210** (FIG. **2**) from absorbing the respective moment load or radial load, which inhibits damage to the static structure **210** from moment or radial loads.
(45) In this document, relational terms such as first and second, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. Numerical ordinals such as “first,” “second,” “third,” etc. simply denote different singles of a plurality and do not imply any order or sequence unless specifically defined by the claim language. The sequence of the text in any of the claims does not imply that process steps must be performed in a temporal or logical order according to such sequence unless it is specifically defined by the language of the claim. The process steps may be interchanged in any order without departing from the scope of the invention as long as such an interchange does not contradict the claim language and is not logically nonsensical.
(46) While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth in the appended claims and the legal equivalents thereof.

Claims

1. A bearing assembly, comprising: an inner race configured to receive a rotating component having an axis of rotation; at least one bearing element coupled to the inner race, the at least one bearing element configured to rotate with the inner race along the axis of rotation; an outer race coupled to the at least one bearing element; and a bearing housing coupled to the outer race and configured to be coupled to a static structure, the bearing housing including a first frangible portion configured to release upon receipt of a moment load greater than a moment threshold and a second frangible portion configured to release upon receipt of a radial load greater than a radial threshold.
2. The bearing assembly of claim 1, wherein the first frangible portion is defined on the bearing housing along an axis substantially perpendicular to the axis of rotation.
3. The bearing assembly of claim 1, wherein the second frangible portion is defined on the bearing housing along an axis substantially parallel to the axis of rotation.
4. The bearing assembly of claim 1, wherein the first frangible portion includes at least one first groove that defines a reduced thickness.
5. The bearing assembly of claim 4, wherein the second frangible portion includes at least one second groove that defines a second reduced thickness.
6. The bearing assembly of claim 4, wherein the at least one first groove is continuous about a perimeter of the first frangible portion.
7. The bearing assembly of claim 4, wherein the at least one first groove comprises a pair of opposed grooves that cooperate to define the reduced thickness.
8. The bearing assembly of claim 1, wherein the first frangible portion includes a plurality of first slots spaced apart about a perimeter of the first frangible portion.
9. The bearing assembly of claim 8, wherein the second frangible portion includes a plurality of second slots spaced apart about a second perimeter of the second frangible portion.
10. The bearing assembly of claim 9, wherein each second slot extends along a major axis, and the major axis is substantially parallel to the axis of rotation.
11. The bearing assembly of claim 8, wherein each first slot extends along a major axis, and the major axis is substantially perpendicular to the axis of rotation.
12. The bearing assembly of claim 1, wherein the bearing housing includes a first coupling section to be coupled to the outer race, a second coupling section configured to be coupled to the static structure, and the first frangible portion and the second frangible portion are defined between the first coupling section and the second coupling section.
13. The bearing assembly of claim 12, wherein the first frangible portion extends radially outward from the first coupling section, and the second frangible portion extends axially from the first frangible portion relative to the axis of rotation.
14. The bearing assembly of claim 13, wherein the second frangible portion is coupled to a first end of the second coupling section, and the second coupling section defines a flange at a second end, with the second end opposite the first end.
15. The bearing assembly of claim 1, wherein the rotating component is a shaft associated with a gas turbine engine.
16. A bearing assembly for a gas turbine engine, comprising: an inner race configured to receive a rotating component having an axis of rotation; at least one bearing element coupled to the inner race, the at least one bearing element configured to rotate with the inner race along the axis of rotation; an outer race coupled to the at least one bearing element; and a bearing housing coupled to the outer race and configured to be coupled to a static structure, the bearing housing including a first frangible portion and a second frangible portion, the first frangible portion extending along a first axis substantially perpendicular to the axis of rotation, the first frangible portion configured to release upon receipt of a moment load greater than a moment threshold, the second frangible portion extending along a second axis substantially parallel to the axis of rotation and the second frangible portion is configured to release upon receipt of a radial load greater than a radial

threshold.

17. The bearing assembly of claim 16, wherein the first frangible portion includes a pair of opposed grooves that cooperate to define a reduced thickness and each groove of the pair of opposed grooves is continuous about a perimeter of the first frangible portion.

18. The bearing assembly of claim 16, wherein the first frangible portion includes a plurality of first slots spaced apart about a perimeter of the first frangible portion and the second frangible portion includes a plurality of second slots spaced apart about a second perimeter of the second frangible portion.

19. The bearing assembly of claim 18, wherein each second slot extends along a major axis, the major axis is substantially parallel to the axis of rotation.

20. The bearing assembly of claim 18, wherein each first slot extends along a major axis, and the major axis is substantially perpendicular to the axis of rotation.